



ActiveHeal

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Abstract

Rehabilitation programs often face challenges such as low patient adherence, limited follow-up, and insufficient communication between patients and therapists. To address these issues, ActiveHeal was designed as a mobile-based rehabilitation support system that assists patients throughout their recovery process. The system includes features such as exercise tracking, video-guided exercises, progress monitoring, reminders, therapist communication, and AI-based assistance within a user-friendly interface. The prototype was developed in Figma following Human-Centered Design principles and evaluated using heuristic evaluation and usability testing with seven participants. The evaluation results indicated that the system provides a clear, accessible, and effective user experience while also identifying areas for further usability improvement. Overall, ActiveHeal demonstrates the potential of mobile rehabilitation systems in enhancing patient engagement and supporting rehabilitation management.

Introduction

Adherence to physiotherapy programs remains a major challenge in rehabilitation care, particularly during home-based recovery where supervision and therapist communication may be limited [1]. Difficulties in remembering exercises and monitoring rehabilitation progress may negatively affect treatment adherence and recovery outcomes.

To address these challenges, this project presents ActiveHeal, a mobile-based rehabilitation support application designed to strengthen continuity of care during rehabilitation. The system supports patient follow-up through rehabilitation guidance, progress monitoring, reminders, and therapist communication.

Unlike traditional rehabilitation follow-up methods that mainly rely on periodic clinical visits, ActiveHeal provides continuous digital support between appointments. Preliminary usability evaluation demonstrated intuitive navigation and improved communication between patients and therapists. The proposed system aims to encourage more consistent treatment adherence, strengthen patient engagement, and enhance continuity of care during rehabilitation.

Methodology

ActiveHeal was developed using an Iterative Human-Centered Design (HCD) approach to optimize physiotherapy adherence. The process began with translating user requirements into the conceptual sketches and low-fidelity wireframes visualized in Figure 1, establishing the core navigation flow.

By leveraging Figma, the prototype evolved into a high-fidelity system. As shown in Figure 2, key features were implemented to enhance the user experience, including automated identity synchronization and real-time exercise timers with repetition counters. To validate the design, usability testing was conducted with seven participants. As illustrated in Figure 3, the empirical data confirmed high task completion rates and user satisfaction across both controlled and dynamic environments, ensuring the system's efficiency and ease of use.



Fig. 1: Low-fidelity conceptual sketches.

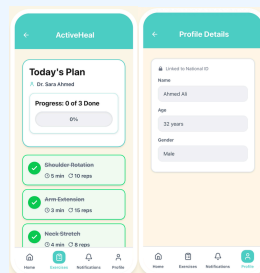


Fig. 2: Core interactive interfaces and features.

FEATURE	USER FEEDBACK
Ease of Use Clarity and overall experience of using the system	Users found the system intuitive, simple, and easy to use with minimal learning effort. VERY HIGH
Navigation Moving between sections and accessing features	Smooth and consistent navigation throughout the application. HIGH
Progress Tracking Monitoring progress and visualizing improvements	Users appreciated the clarity in progress tracking and performance visualization. HIGH
Treatment Adherence Support Support for exercise adherence and reminders	Reminders and structured plans helped users stay consistent with their exercises. HIGH
Therapist Communication Interaction and communication (with therapist / AI)	Generally positive, but some users had difficulty distinguishing between AI and therapist responses. MODERATE-HIGH

Fig. 3: Usability testing performance metrics.

Results and Discussion

Usability testing showed that participants were able to complete key tasks successfully, indicating that the system is generally easy to use and supports rehabilitation activities effectively [4]. Users reported that progress tracking, reminders, and therapist communication improved treatment adherence. In addition, navigation between the main sections was smooth and understandable for most participants. However, some users initially experienced difficulty distinguishing between AI-generated responses and therapist communication. Limited feedback after task completion was also identified during testing. These issues were improved through clearer interface labeling and enhanced progress indicators. As shown in Fig. 4, the system supports exercise completion and rehabilitation progress tracking through interactive interfaces and real-time feedback.

Key Findings

1. Successful task completion by most participants
2. Improved treatment adherence through reminders and tracking
3. Clear and simple navigation experience
4. Enhanced patient-therapist communication
5. Increased user engagement during rehabilitation tasks

Discussion

The findings highlight the importance of user-centered rehabilitation systems in improving patient engagement and continuity of care [3]. Features such as progress tracking, reminders, and structured exercise plans contributed to a more organized rehabilitation experience and supported better adherence to physiotherapy programs. As shown in Figs. 5 and 6, additional features supported patient monitoring, communication, and rehabilitation engagement. The results also align with previous rehabilitation studies emphasizing the role of digital healthcare technologies in enhancing communication, motivation, and treatment monitoring [1], [2]. Furthermore, the simplicity and consistency of the interface reduced cognitive effort and improved usability during interaction. Overall, the findings demonstrate the potential of ActiveHeal to provide a more accessible, supportive, and effective rehabilitation experience.

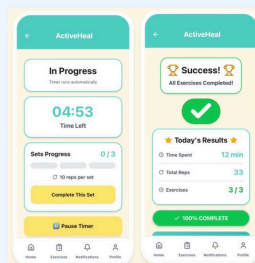


Fig. 4: Exercise completion and rehabilitation progress tracking interfaces.

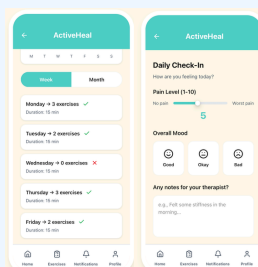


Fig. 5: Daily check-in and progress tracking interfaces.

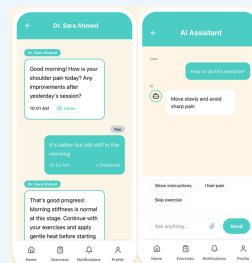


Fig. 6: AI assistance and therapist communication interfaces.

Conclusion

ActiveHeal demonstrated the potential of user-centered rehabilitation systems in supporting physiotherapy patients and therapists throughout the rehabilitation process. The evaluation findings showed that the system is generally easy to use and supports rehabilitation tasks effectively through features such as progress tracking, reminders, and therapist communication. Usability testing and heuristic evaluation also helped identify interface improvements related to feedback clarity and interaction design. Future work may include developing a fully functional implementation of the system and evaluating it with a larger group of users in real-world clinical environments.

References

